

Elana Wong
Dr. Devon Gorry
ECON 8700
13 December 2024

What Impacts Your Relaxation?

INTRODUCTION

My question of this analysis is what factors are most strongly associated with total relaxation and leisure time. I will use a regression analysis to predict total relaxation and leisure time, including personal activities. Considering the variables that I will show in my regression such as, work hours, gender, marital status and whether or not one has young children, I want to find out which factor of life has a greater influence on self reported leisure time. I also want to see the impact that being married, having a kid, and being female has on your relaxation time because everyone says that moms never get a day off. Unfortunately I find that a large majority of my interesting variables are not statistically significant.

LITERATURE REVIEW

Zhao's article, titled "From Motherhood Premium to Motherhood Penalty? Heterogeneous Effects of Motherhood Stages on Women's Economic Outcomes in Urban China", focuses on the impact of children's age and the relationship between children's age and mother's salary specifically focused on China. Though this may not exactly interact with the relaxation hours, it is associated with my topic as it begins to provide context to support my hypothesis that maybe women tend to be held more responsible for the caregiving responsibilities of the children and therefore it impacts their wages or labor force participation. This article is particularly important as it discusses the impact on women's economic outcomes as they balance work with child rearing obligations. The data used in this article was sourced from the China Health and Nutrition Survey (CHNS). The survey used captured family changes between 1989 to 2011 and is a longitudinal study, which differs from my data which is cross sectional, because it follows the same households over time. They also focus specifically on urban women so they excluded rural statistics and included all women, working or not, in the data. Zhao does use a person-fixed-effects model in order to control for the within-person change in labor force participation and income over time with the development of the child's independence and growth. The main analysis was conducted to study the outcome of women's labor force participation and income.

The conclusion of this study is that overall, women's labor force participation declined after 1990, which is most likely attributed to change in societal expectations, and that mother's tend to base their labor force participation off of the needs of their children. For example, a lower labor force participation was reported for mothers who are parents of young children aging from

0-3 years old which, hypothetically, is caused by the fact that women tend to change their economic decisions based on the demands of their families. There was also, on average, a higher income for women with school-aged children than that of any other mother with other aged children.

According to Lozano in her article titled, “Non-Standard Work Schedules, Gender, and Parental Stress,” the researchers seek to examine the association between non-standard work schedules and stress levels among dual-income earning adults with children. Though this may not be exactly what my analysis looks like, I feel as though parental stress, as researched in this article, and my analysis of what aspects of life, especially work hours and parental responsibilities, impact your relaxation hours are appropriately inversely related so as stress increases, relaxation decreases. Hypothetically as your stress levels increase, your ability to relax likely decreases, which means that even if you have more free time you may forget to spend that time actively relaxing. In the article, the researchers use a sample of 1,932 working parents from the Canadian 2010 General Social Survey, which includes a time-use diary. They estimate the odds of being stressed, and then they estimate the comparison between high vs. medium levels of stress.

As a result of their analysis they found a large discrepancy between the non-standard scheduled working hours and the perceived levels of stress between mothers and fathers. For mothers, having a non-standard schedule allows them less stress. The researchers tested the hypothesis that the association between non-standard hours and stress is mediated by the flexibility and predictability of the work schedule. The results were not statistically significant in the male sample but they did suggest that fathers with non-standard schedules are more stressed than those with standard working hours and the opposite is true for mothers. The author did note that this is likely because mothers feel more pressure to be more available for their families and therefore plan to have jobs with non-standard hours in order to be present for their children whereas fathers typically do not follow that way of thinking.

Looze’s article titled, “Young Women's Job Mobility: The Influence of Motherhood Status and Education,” discusses the implication of having children and the impact on women’s income. This article is least related to my research question, but it has some of the same variables such as education, income, and whether or not the women have children. Since this article is using the same variables, despite being used in different ways, it is still important to have in my analysis. The control variables include work experience, tenure, school enrollment, annual weeks worked, weekly hours worked, and self-employment. The dependent variables in their models are the natural logarithm of the hourly wage of the respondent's mothers at some point during the observation current or most recent job. Looze’s article focuses more on the movement of women in and out of the labor force and job separations. The author acknowledges that motherhood is not always the reason for job separations, but the author did discover that women in their 20s fall victim to a larger wage discrepancy than women who delay childbirth till their 30s. Looze also investigated that for childless versus motherhood status, the wage gain difference is

approximately five percent less for mothers. These differences contribute to the motherhood wage penalty but more so for mothers who have lower education levels.

DATA DESCRIPTION

My data is from Kaggle which is an online data sharing site. It is called “Sleep Patterns” and when I initially downloaded it, it had 34 different variables but I wanted to focus on just a couple interesting ones. After cleaning, my dataset has 706 observations with 7 variables. I had age, which is a continuous numerical variable that is age in years and in my dataset, ranges from 23 to 65 years old. Yngkid is a dummy variable that is equal to 1 if the respondent has a child between the ages of 0 and 3, it does not count how many, for instance if someone were to have a 2 year old and a newborn. The range for the young kid variable is from 0 to 1 but the mean is 0.1289 meaning that there is a large margin of respondents who do not have children between 0 and 3 years old. Education is education measured in years and its overall years of education. The range is from 1 to 17 years which is likely a master’s degree at the highest and then kindergarten at the lowest. It is important to note for this variable that the mean is 12.78 meaning that there is a large margin of respondents likely to have only a high school diploma, which is typically what people associate with 12 years of education. There is a dummy variable on gender, male, that is equal to 1 if the respondent is a male and 0 otherwise. There is a dummy variable on marriage, marr, that is equal to 1 if the respondent is married and 0 otherwise. This does not account for people who have been married but the mean value of marriage is 0.82 meaning that a large portion of the dataset is married. Total work is the total minutes spent working each week. The mean is 2123 which is around 35 hours per week meaning that a lot of the respondents probably do not work full time jobs which you would then expect to see 40 hrs per week or in our values 2400 minutes per week. Rlxall or relax all is a numerical variable in minutes per week for total relaxation and leisure time, including personal activities.

I chose this subset of data specifically because these are the variables that I have a hypothesis on that are going to be the most explanatory. There were other variables like Black, construction worker, and Protestant, but because there were not categorical variables for other variables like White, Asian, doctor, teacher, Catholic or Baptist, I did not feel like the other variables would be of any use.

rlxall	totwrk	age	educ	male	marr	yngkid
Min. :1380	Min. : 0	Min. :23.00	Min. : 1.00	Min. :0.0000	Min. :0.0000	Min. :0.0000
1st Qu.:3150	1st Qu.:1554	1st Qu.:29.00	1st Qu.:12.00	1st Qu.:0.0000	1st Qu.:1.0000	1st Qu.:0.0000
Median :3428	Median :2288	Median :36.00	Median :12.00	Median :1.0000	Median :1.0000	Median :0.0000
Mean :3438	Mean :2123	Mean :38.82	Mean :12.78	Mean :0.5666	Mean :0.8215	Mean :0.1289
3rd Qu.:3720	3rd Qu.:2692	3rd Qu.:48.00	3rd Qu.:16.00	3rd Qu.:1.0000	3rd Qu.:1.0000	3rd Qu.:0.0000
Max. :6110	Max. :6415	Max. :65.00	Max. :17.00	Max. :1.0000	Max. :1.0000	Max. :1.0000

It is important to note that the amount of relaxation time has a very large spread from 1380 to 6110 minutes. This can also be seen with total time worked per week, with the spread from 0 to 6415 minutes, and with education which ranged from 1 to 17 years. There is a larger sample of people who are married as we can see in the married column because the mean is

closer to 1. There is also a large sample of people who do not have young children as we can see in the *yngkid* spread where the mean is much closer to 0.

In Kaggle there is a search filter you can add which allows for you to only search through data sets that do not have any missing variables. So when I started with my data I had no NAs and no unexplained zeros. An explained zero would be a male dummy where the value is between 0 and 1 and the zero means female versus age is 0 which means the value must be missing. This just made my data “cleaning” nonexistent essentially. I did however create a subset in R in order to focus on specific variables. This plot shows the relationships between work time and relaxation. There is a lot of noise towards the middle but there is a vague downward trend which is expected. It is expected that since your time is limited everyday that using your time to work would then decrease the amount of time that you would be spending relaxing instead. An increase in one usage of your time definitely results in a decrease in the other usage of your time, i.e. relaxing versus working.

METHODS

I will be running three separate regressions. First I want to run a regression that analyzes the determinants of relaxation and leisure time. This will be a baseline regression which I will use to build the other regressions off of.

Regression 1: Baseline of Total Work Hours on Relaxation Time.

$$rlxall = \beta_0 + \beta_1 totwrk + u$$

The next regression I will run includes being married or having children and how that impacts your relaxation time. I hypothesize that being married has a positive effect on your relaxation and leisure time per week and that having a young kid has a negative effect on your relaxation and leisure time per week. Regression 2: $\beta_2, \beta_3 < 0$: Being Married or Having Young Children Reduces Relaxation Time.

$$rlxall = \beta_0 + \beta_1 totwrk + \beta_2 marr + \beta_3 yngkid + u$$

The last regression will include an interaction term ($marr \times yngkid$), which captures the combined effect of being married and having young children. If $\beta_7 > 0$, then there is a positive interaction between being married and having a young kid. This means that the combination of these two factors has a stronger effect on relaxation time than the individual coefficients that β_5 and β_6 have on relaxation time.

$$rlxall = \beta_0 + \beta_1 totwrk + \beta_2 age + \beta_3 educ + \beta_4 male + \beta_5 marr + \beta_6 yngkid + \beta_7 (marr \times yngkid) + u$$

This combination of regression analysis will allow me to see the interaction and impact that being married and having a young kid truly has on leisure and self reported relaxation time.

RESULTS

Surprisingly, a lot of the variables are not statistically significant. Total time worked per week in units of minutes is statistically significant which is expected because relaxation and working are different ways to use your time so it makes sense.

Each one unit increase in age is associated with a 2.932 minute increase in relaxation time, *ceteris paribus*. This value is statistically significant at the 10% significance level. Education is statistically significant at the 5% significance level. Each one unit increase in education is associated with a 16.730 minute decrease in relaxation time, *ceteris paribus*. This is likely due to the fact that people who pursue higher education tend to be more hard working, more dedicated, and more likely to be working on higher level jobs that require lots of time and effort. It is important to note that the constant is in units of minutes per week not hours, because approximately 3,800 hours would be absurd, even if you are unemployed and have nothing to do. It is also impossible. The last statistically significant variable that is interesting is male. Being male is associated with a 93.845 increase in minutes of relaxation per week, *ceteris paribus*. This is statistically significant at the 5% significance level.

Regression Results			
Dependent variable:			
	(1)	rlxall (2)	(3)
totwrk	-0.196*** (0.019)	-0.196*** (0.019)	-0.210*** (0.021)
age			2.932* (1.753)
educ			-16.730** (6.770)
male			93.845** (40.634)
marr		22.098 (48.268)	2.761 (49.464)
yngkid		-77.236 (55.141)	254.931 (246.137)
interaction			-310.957 (251.964)
Constant	3,854.978*** (44.944)	3,845.791*** (60.391)	3,934.577*** (138.232)
Observations	706	706	706
R2	0.128	0.130	0.153
Adjusted R2	0.126	0.126	0.144
Residual Std. Error	486.417 (df = 704)	486.404 (df = 702)	481.370 (df = 698)
F Statistic	103.050*** (df = 1; 704)	35.030*** (df = 3; 702)	18.009*** (df = 7; 698)
Note: *p<0.1; **p<0.05; ***p<0.01			

The interaction term on this is the combined effect of being married and having young children and it is a negative coefficient. This states that being married and having a child is associated with a 310.957 minute decrease in relaxation time per week which is expected because you would expect that being married and having a child would cause you to have less time for relaxation. There is a negative interaction, meaning the combination of being married and having a child under three reduces the time for relaxation and leisure relative to the sum of their separate effects, the coefficients on marriage and on having a child between zero and three years old. The coefficient on marriage is positive and indicates that being married is associated with a 2.761 minute increase in relaxation per week, *ceteris paribus*. The coefficient on young children between the ages of zero and three is positive and indicates that having a child between zero and three is associated with a 254.931 minutes increase in relaxation time per week, *ceteris paribus*. It is strange that the coefficient on young children is positive because you would not think that having a child would increase your time for relaxation every week but then again this coefficient is not statistically significant.

The adjusted Rsquared is pretty low on this therefore telling us that the variables that are included in this regression does not explain the dependent variable very well, so working hours, age, being married, and having a child are not very explanatory variables as to how much people are able to relax each week. I feel like I would have done better if I had a stress variable that was able to calculate the level of stress of a person and their relaxation hours per week. The large

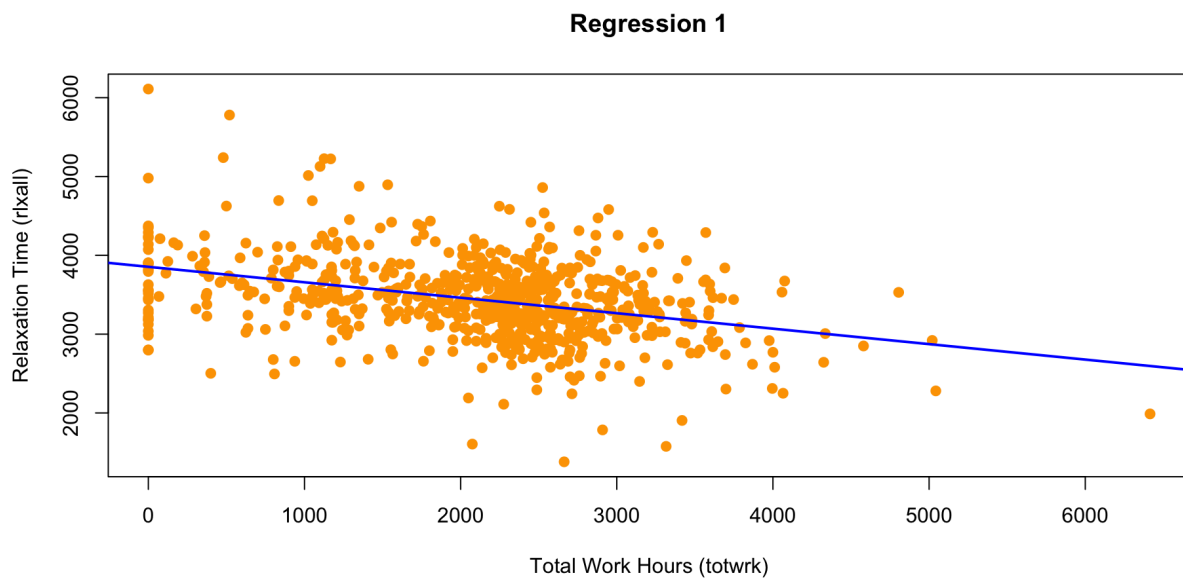
	rlxall	totwrk	age	educ	male	marr	yngkid
rlxall	1.00000000	-0.357333899	0.11726585	-0.11966714	-0.06317110	0.01850142	-0.050709256
totwrk	-0.35733390	1.00000000	-0.04956989	0.03260422	0.37645846	-0.02512196	0.008818588
age	0.11726585	-0.049569891	1.00000000	-0.26583129	0.03193915	0.01755770	-0.319697898
educ	-0.11966714	0.032604217	-0.26583129	1.00000000	0.03576439	-0.03278489	0.104781855
male	-0.06317110	0.376458464	0.03193915	0.03576439	1.00000000	0.16712137	0.106131271
marr	0.01850142	-0.025121964	0.01755770	-0.03278489	0.16712137	1.00000000	0.135131544
yngkid	-0.05070926	0.008818588	-0.31969790	0.10478186	0.10613127	0.13513154	1.00000000

standard error helped me recognize that it is likely these variables are not helpful in determining the true relationship between relaxation hours and all these lifestyle factors.

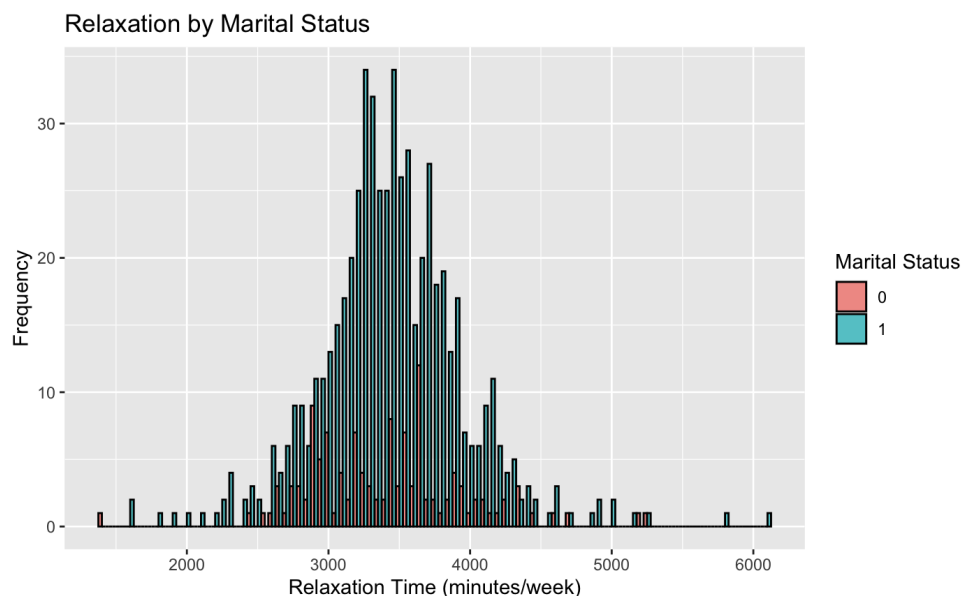
This simple correlation matrix tells us the strengths of relationships between variables and ideally we would want variables that are close to one or negative one because this shows either a strong positive or negative correlation. The “strongest” we get is around ~0.4 which is not very strong in terms of correlation, we want it to be further away from zero and closer to the absolute value of one.

RESULTS – PLOTS

Moving onto my regressions, Regression 1 is the simple baseline regression on relaxation time per week and total work per week. I put it in a scatter plot so we can see the noise that is centered around the 2000 minute mark for working per week. We can also see the downward trend. A large majority of the data is between 1000 and 4000 minutes per week. The data looks pretty heteroskedastic, you can see the large error towards low total work hours and then as total work hours increase you can see the error decrease which is in line with a heteroskedastic regression.



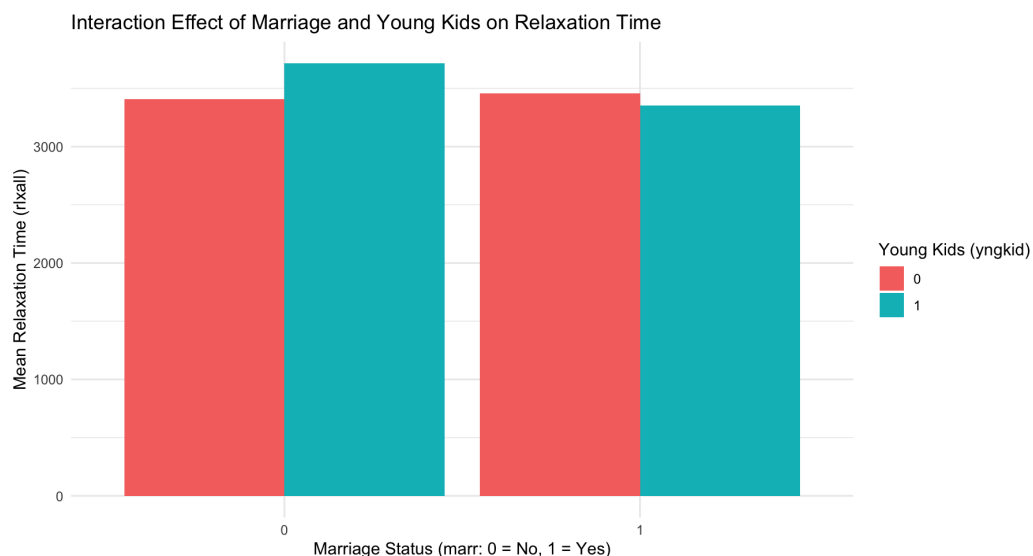
For Regression 2, I decided to show it on a histogram because that's where you can see the relaxation time for married versus unmarried people which I thought was interesting because though the dataset has significantly more married data points, married is has a normally distributed relaxation time of minutes per week and it is very difficult to visualize a normal distribution for the unmarried respondents in the dataset.





Another way I wanted to visualize the model was through a scatter plot, this is definitely more difficult to read. Since the values on married and young kids are not statistically significant you cannot really see a strong difference between the linear model with just relaxation on total work hours, and the linear model with marriage and young kids fixed.

Lastly for Regression 3 that has the interaction term of the two dummy variables of married and young kids, I decided to go with a bar chart because I feel like it visualized the data well. You can see the mean relaxation time for someone who has a young kid between the ages 0 and 3 and is married has the lowest mean relaxation time per week, they are represented by the blue bar on right. Next you can see that people who are not married and do not have young kids have a slightly higher mean and they are represented by the red bar on the left. People who are married and do not have children have an even higher mean than people who are not married. Now finally, people who have young kids and are not married have the highest mean relaxation



time per week. This is confusing and a bit backwards as you would expect people who are raising a young child by themselves would have the lowest amount of relaxation time. It is important to recognize that married, young kid and the interaction term were all not statistically significant at any level and that the R squared showed that these variables are not very explanatory to helping explain variation in the model. The low adjusted R squared tells us that this regression model does not do well at explaining a larger proportion of the variability in my data, signifying a poor fit to the data. So even though I included this histogram it is not helpful in trying to understand the data as much because this data is not a good fit for an explanation of the independent variable of relaxation time per week.

LIMITATIONS AND EXTENSIONS

Do the assumptions hold? Let's see. In regards to being a random sample, I would say this is a random sample according to the description of the dataset on the website, Kaggle, so to the extent of my knowledge, this is random sampling. In regards to linear in parameters, I believe that all of these variables have a constant impact on the relaxation and leisure time variable and I

```

studentized Breusch-Pagan test
data: regression1
BP = 5.0032, df = 1, p-value = 0.0253

studentized Breusch-Pagan test
data: regression2
BP = 5.8709, df = 3, p-value = 0.1181

studentized Breusch-Pagan test
data: regression3
BP = 10.671, df = 7, p-value = 0.1536

```

do not believe that they have diminishing or increasing effects or that the marginal effect on them changes. There is variation in X when you look at the summary tables. There is variation in ages, education, marital status, offspring count, and minutes worked per week. I did explain earlier that there are substantially more married people in the dataset who do not have young children between ages zero and three as explained in the summary table with the mean of each of those dummy variables being .82 and 0.12 respectively. In terms of zero conditional mean, there are definitely unobserved variables that are in the error term that could definitely impact the relaxation time. I am thinking of categorical variables like what specific field someone is in and along with that, their respective stress level. I am assuming that someone who works on Wall Street is probably more stressed than an average elementary school teacher. Health status is another unobserved variable in my regression that could have an impact on the relaxation time which means that the condition of zero conditional mean is definitely violated. In terms of bias, I would say that reverse causality in endogeneity could definitely be something that could bias this. A higher relaxation time results in people working less, so I can definitely see how that would cause an upward bias. However I think the easiest bias to get rid of would be the omitted variable bias that most likely biases downwards such as higher stress levels which is positively correlated with work time and negatively correlated with relaxation time.

In terms of whether or not the model is homoskedastic, the Bruesch Pagan test shows us the P-value and unfortunately for Regression one the P-value = 0.0253 which means it rejects the null hypothesis and therefore is heteroskedastic. However for Regressions two and three, the P-values are greater than 0.05, which fails to reject the null hypothesis, indicating that they are homoskedastic. In summary, for limitations, we could definitely benefit from having more

related and hypothetically causal variables like stress level, spouse income level, spouse employment, and job type.

CONCLUSIONS

I learned that it is important to know where you are getting in terms of data and that there are a lot of omitted variables that you can think of when you are running an analysis and you do not get the result that you want. This also has helped me remember the importance of statistical significance and that even though you run a regression about something, it does not always mean that you are including a relevant variable. My lack of statistically significant results is causing me to believe that I probably did not have the correct variables in order to conduct the analysis I wanted to. If I were to redo this again, I would definitely search for more categorical variables like types of jobs and stress level. I feel as though when people are perpetually stressed out they tend to underestimate how much time they spend relaxing each week. In terms of actual statistical results, it is important to understand the data that you are running, like what variables you do and do not have, that can maybe help explain why you didn't get the result that you wanted or expected. I understand now, after looking at the regression summary and the correlation coefficient that the plots reasonably do not make a whole lot of sense because I am not using variables that are explaining the data super well.

Works Cited

- Looze, Jessica. "Young Women's Job Mobility: The Influence of Motherhood Status and Education." *Journal of Marriage and Family*, vol. 76, no. 4, 2014, pp. 693–709. *JSTOR*, <http://www.jstor.org/stable/24582740>. Accessed 1 Dec. 2024.
- Lozano, Mariona, et al. "Non-Standard Work Schedules, Gender, and Parental Stress." *Demographic Research*, vol. 34, 2016, pp. 259–84. *JSTOR*, <http://www.jstor.org/stable/26332035>. Accessed 2 Dec. 2024.
- Zhao, Menghan. "From Motherhood Premium to Motherhood Penalty? Heterogeneous Effects of Motherhood Stages on Women's Economic Outcomes in Urban China." *Population Research and Policy Review*, vol. 37, no. 6, pp. 967–1002. EBSCOhost, <https://doi-org.libproxy.clemson.edu/10.1007/s11113-018-9494-0>. Accessed 1 Dec. 2024.